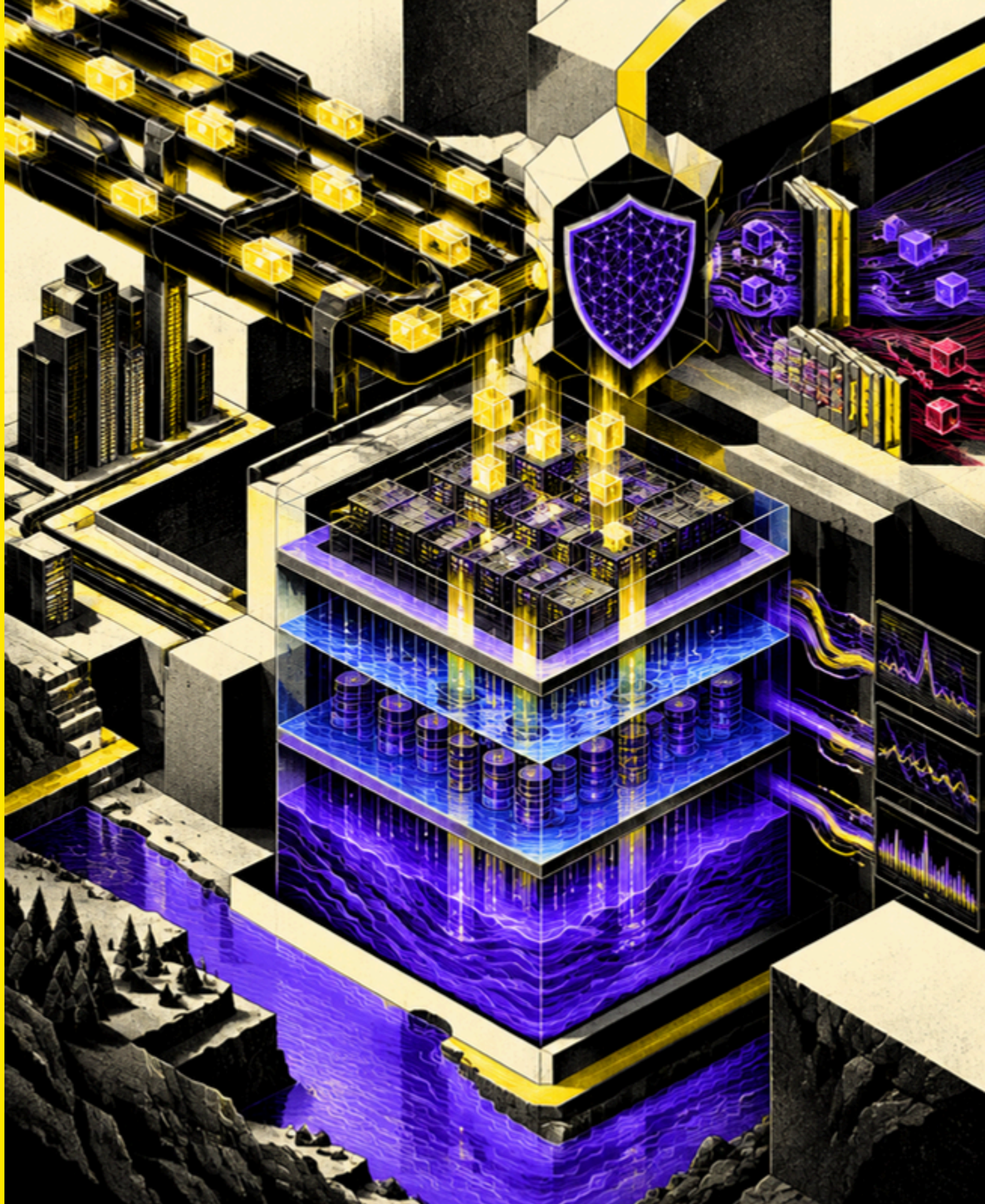
ENTERPRISE STREAMING / ANTI-FRAUD

SIC8 REAL-TIME PAYMENTS

Zero DB inserts. Sub-10ms scoring. One governed lakehouse.



Velocity without contention

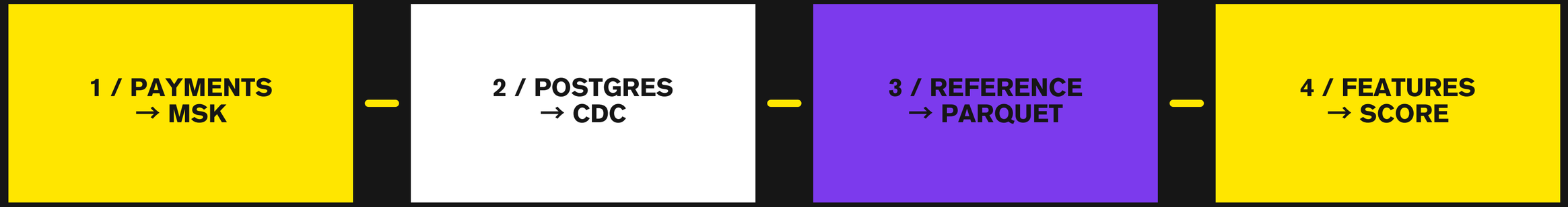
The platform treats database writes as a scalability tax—and removes them from the payment hot path.



INVARIANTS • replayable events • idempotent processing • no target leakage • zero data loss

Four paths. One governed truth.

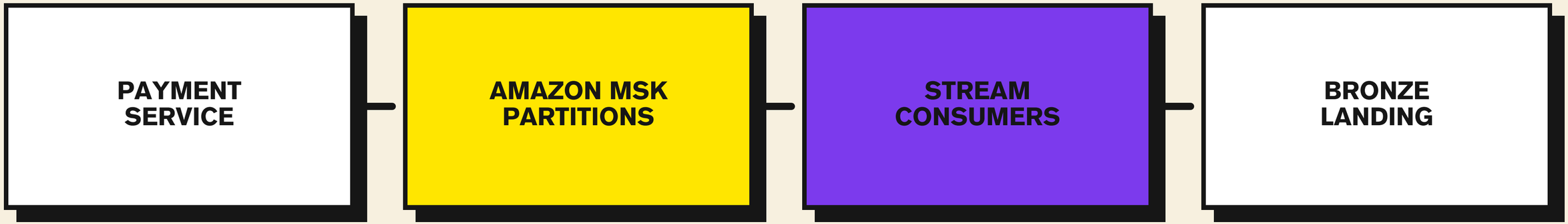
Streaming, CDC, batch and inference converge without coupling the payment authorization path.



S3 + DELTA LAKEHOUSE

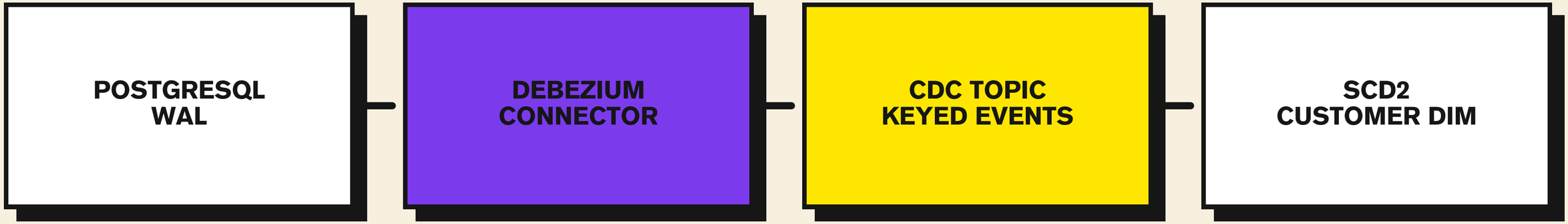
Path 1 / Write events, not rows

The transaction service publishes directly to managed Kafka. Partitioning unlocks horizontal scale; consumers remain independently deployable.



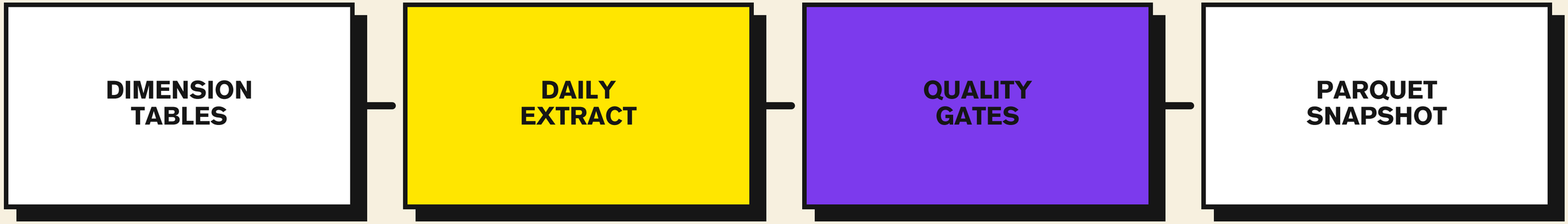
Path 2 / Read the WAL once

Debezium captures customer and account mutations from PostgreSQL logs—no application double-write, no polling loop.



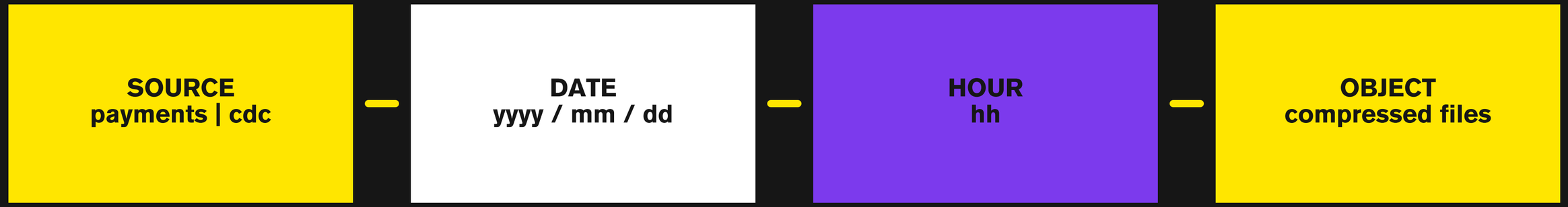
Path 3 / Batch, made deterministic

Daily reference snapshots are extracted, validated and encoded as columnar Parquet for fast joins and reproducible backfills.



Bronze is an immutable contract

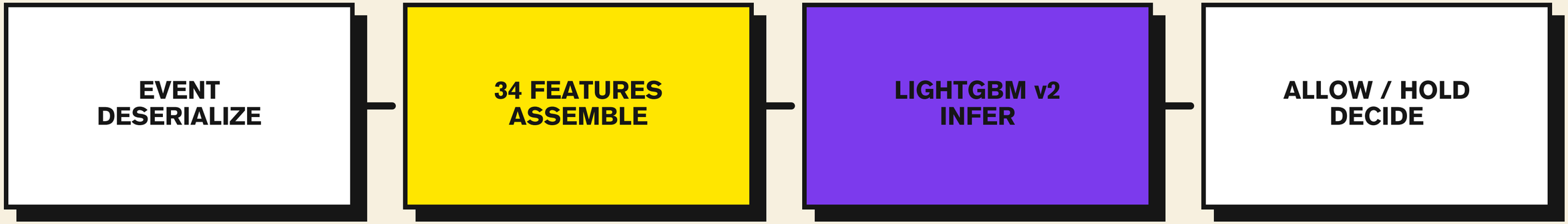
Partitioning balances write throughput, replay precision and downstream scan efficiency.



RAW ≠ MESSY • append-only • schema stamped • lineage preserved

Decision before authorization

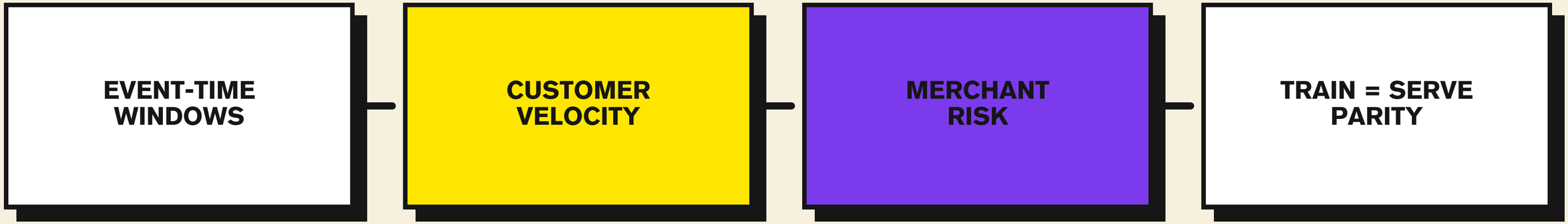
A lean LightGBM v2 inference path evaluates each payment against hydrated behavioral state in under ten milliseconds.



< 10 ms

34 features. One definition.

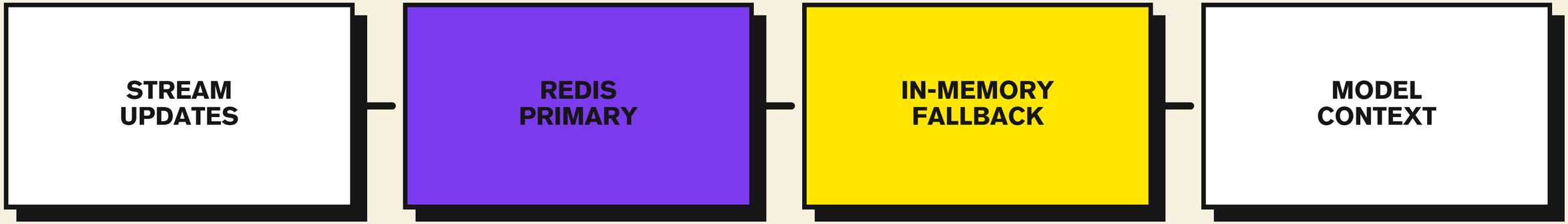
The zero-leakage contract makes offline training and online scoring mathematically identical.




```
same source  +  same clock  +  same defaults  +  same transforms
```

Fast state, resilient state

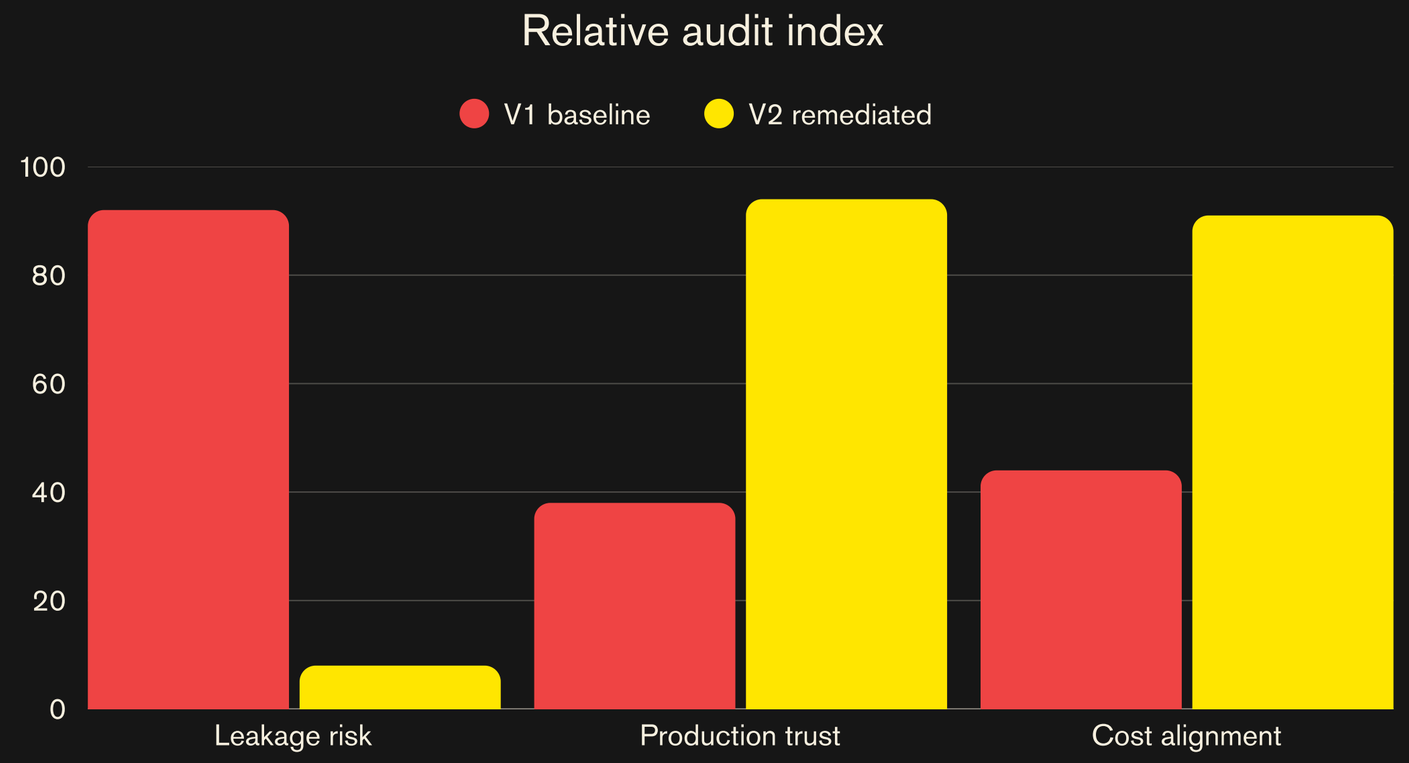
Redis is the primary online feature store; a thread-safe local cache absorbs outages without stopping authorization.



SUB-MILLISECOND HYDRATION / BOUNDED STALENESS / ZERO-DOWNTIME FALLBACK

V2 optimizes the cost of being wrong

Leakage remediation trades inflated offline confidence for trustworthy production decisions.

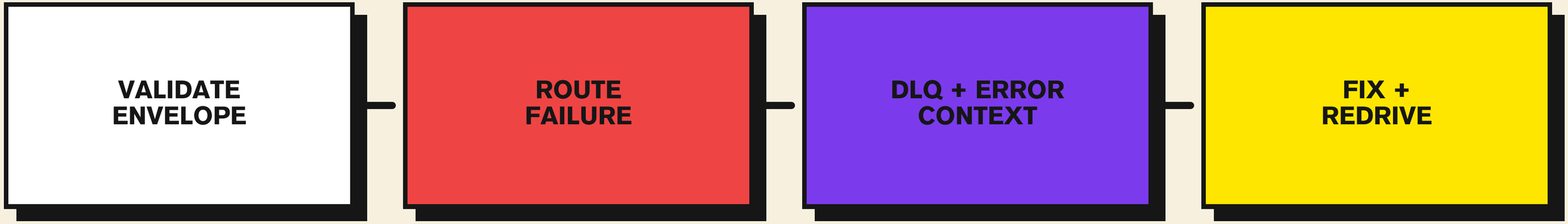


NO LEAKAGE

Empirical thresholds tuned to the banking cost curve—not vanity accuracy.

Failure becomes a replayable event

Malformed payloads, poison pills and contract violations are isolated with full context—never silently dropped.



Medallion turns events into evidence

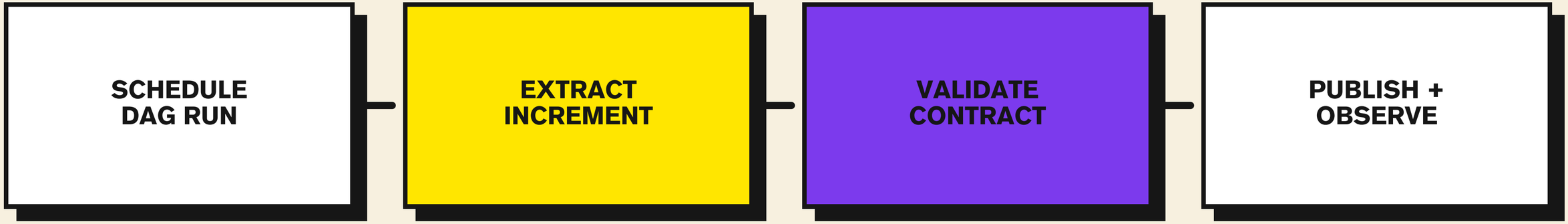
Continuous Delta processing adds quality, identity and business meaning without erasing the source record.



`Delta Lake` • `checkpoints` • `MERGE semantics` • `refund matching` • `SCD Type 2`

Orchestration is control, not compute

Airflow coordinates incremental extracts, quality gates and infrastructure tasks while preserving idempotent retries.



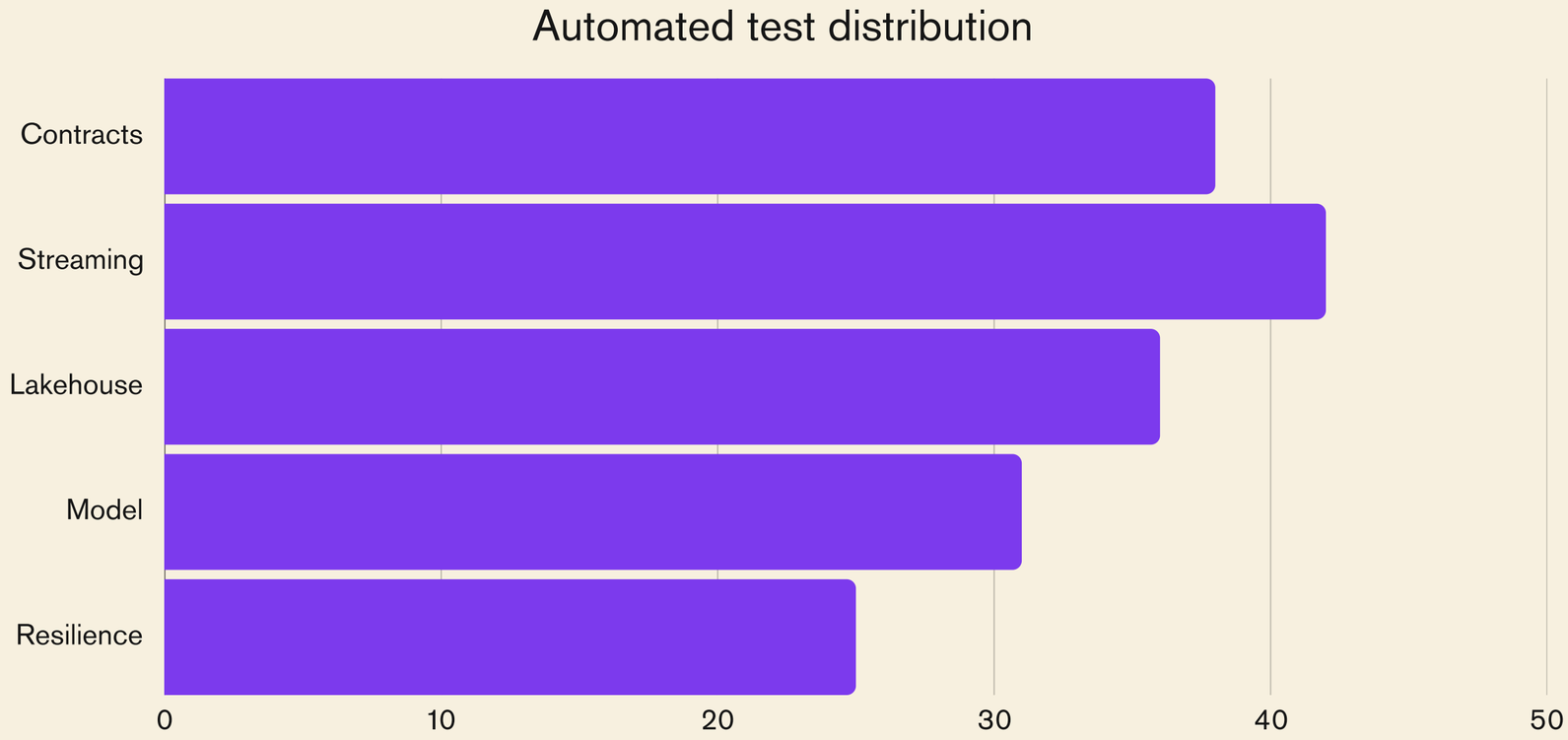
Trust boundaries are explicit

Every layer is encrypted, identity-scoped and isolated; secrets never enter source code or runtime images.



172 tests—and the cloud evidence

Contracts, transformations, model behavior and stress limits are verified as a single production acceptance system.



172

AUTOMATED TESTS

10–100 TPS
cloud stress envelope

Production-ready by design

A reproducible, observable anti-fraud lakehouse that scales payments without scaling database contention.

STREAM WITHOUT LOCKS — **SCORE BEFORE AUTH** — **REPLAY WITHOUT LOSS** — **GOVERN END TO END**

SIC8 / REAL-TIME PAYMENTS